

4DOTS Landing Page

A fast, dependency-free landing page for 4DOTS, hosted on Cloudflare Pages. Lead submissions are validated by Cloudflare Turnstile and written to a private Google Sheet through a small Google Apps Script web app.

Architecture

```
Visitor
-> Cloudflare Pages static site
-> Google Apps Script web app
    -> Cloudflare Turnstile verification
    -> server-side input validation
    -> private Google Sheet
```

The frontend contains no secret values. The Turnstile secret and Sheet ID stay in Apps Script Properties. See [REQUIREMENTS.md](#) for the tested runtime versions, current sheet schema, and redeployment rules.

Repository Structure

docs/	Public Cloudflare Pages output
index.html	
styles.css	
app.js	
_headers	
apps-script/	Google Apps Script source
Code.gs	
appsscript.json	
tests/	Automated contract and validation tests
requirements.txt	Python dependencies for local tests and CI
REQUIREMENTS.md	Tested runtime, configuration, and deployment
requirements	
.node-version	Node.js version used by CI and compatible tooling
.nvmrc	Node.js version used by nvm
wrangler.jsonc	Cloudflare Pages project configuration
assets/	Local design references; not published from docs/

Run Locally

From the repository root:

```
python3 -m http.server 8000 --directory docs
```

Open <http://localhost:8000>. Turnstile hostname restrictions may prevent the production widget from verifying on localhost; use the production Pages URL for an end-to-end form test.

Run Automated Tests

Create the local virtual environment and install the test dependency:

```
python3 -m venv .venv
.venv/bin/python -m pip install -r requirements.txt
```

Use Node.js 24 for JavaScript checks and Wrangler:

```
nvm use
node --version
```

For Cloudflare builds, add the environment variable `NODE_VERSION=24` under the project's build settings if the detected Node version is older.

Run the suite:

```
.venv/bin/pytest
```

The tests verify the Cloudflare Pages configuration and headers, Apps Script timezone, frontend/backend allowed-value contract, backend validation and formula-injection protection, and JavaScript syntax. GitHub Actions runs the same suite on pushes and pull requests.

Configure the Google Sheet

1. Create a private Google Sheet for leads.
2. Copy the ID from its URL:

```
https://docs.google.com/spreadsheets/d/SHEET_ID/edit
```

3. Do not make the Sheet public. The Apps Script deployment writes to it using the deploying Google account.

The script creates a **Leads** tab and this header row on the first valid submission:

```
Received At | Name | Phone | Email | Services | Industry | Website | Source
| Consent | Status | Submission ID
```

Use a new or empty tab. There is no **Message** column, and the column order must remain aligned with the values written by **Code.gs**.

Configure Cloudflare Turnstile

1. Create a free Turnstile widget in the Cloudflare dashboard.
2. Under **Hostname Management**, add the launch hostname, such as **4dots.pages.dev**, and any custom domain you plan to use. Enter hostnames only: do not include **https://**, paths, ports, or wildcards.
3. Copy the public **site key** and private **secret key**.
4. Put the site key in **docs/app.js**:

```
turnstileSiteKey: "YOUR_PUBLIC_SITE_KEY",
```

Never put the Turnstile secret key in **docs/** or anywhere else served by Cloudflare Pages.

If the widget displays **Unable to verify website**, check the error code shown below the form or in the browser console:

Error	Meaning	Fix
110200	Domain not authorized	Add the exact current hostname under the widget's Hostname Management
110100 or 110110	Invalid or unknown site key	Copy the site key from the same widget again
400070	Site key disabled	Re-enable or replace the widget
200500	Turnstile iframe blocked	Disable blockers and confirm challenges.cloudflare.com can load

The site key in **docs/app.js** and the **TURNSTILE_SECRET** Apps Script property must come from the same Turnstile widget. Also add the same production hostname to the Apps Script **ALLOWED_HOSTNAMES** property.

Deploy Google Apps Script

1. Open [Google Apps Script](#) and create a project.
2. Replace its source with **apps-script/Code.gs**.
3. In **Project Settings**, enable showing the manifest file and replace it with **apps-script/appsscript.json**.
4. Under **Script Properties**, add:

Property	Required	Value
<code>SHEET_ID</code>	Yes	Google Sheet ID
<code>TURNSTILE_SECRET</code>	Yes	Private Turnstile secret key
<code>SHEET_NAME</code>	No	Defaults to <code>Leads</code>
<code>ALLOWED_HOSTNAMES</code>	Recommended	Comma-separated hosts, such as <code>4dots.pages.dev, www.example.com</code>
<code>RETURN_URL</code>	Recommended	Full public landing-page URL

- For the initial deployment, select **Deploy > New deployment > Web app**.
- Set **Execute as** to yourself.
- Set access to **Anyone**.
- Authorize the requested Google Sheets and external-request scopes.
- Copy the deployed `/exec` URL.
- Put that public URL in `docs/app.js`:

```
formEndpoint: "YOUR_GOOGLE_APPS_SCRIPT_EXEC_URL",
```

The `/exec` URL is a public endpoint and is safe to place in the frontend. The receiver still requires a valid, single-use Turnstile token and validates all submitted values server-side.

Redeploy Apps Script

After changing `Code.gs`:

- Save the updated Apps Script files.
- Open **Deploy > Manage deployments**.
- Edit the existing web-app deployment.
- Select **New version** and deploy.

Editing the existing deployment preserves its `/exec` URL, so `CONFIG.formEndpoint` in `docs/app.js` does not need to change.

Only update `CONFIG.formEndpoint` and redeploy Cloudflare Pages when you create an entirely new Apps Script deployment with a new `/exec` URL. Script Property changes such as `SHEET_ID`, `SHEET_NAME`, `ALLOWED_HOSTNAMES`, or `RETURN_URL` normally do not require a new deployment version.

The Apps Script confirmation page uses top-level navigation for its return link. This is required because Apps Script renders HTML inside a sandboxed iframe while Cloudflare Pages correctly blocks iframe embedding.

Deploy Cloudflare Pages

- Push the repository to GitHub.

2. In the Cloudflare dashboard, open **Workers & Pages** and create a Pages application using **Pages > Import an existing Git repository**. Do not create a Workers application; Workers Builds defaults to `npx wrangler deploy`, which is not the Pages deploy command.
3. Connect this repository and use `main` as the production branch.
4. Select no framework preset, leave the build command blank, and use `docs` as the build output directory.
5. Deploy the project. Cloudflare will publish future pushes automatically and create preview deployments for non-production branches.

The checked-in `wrangler.jsonc` keeps the Pages project name, output directory, and compatibility date in source control. Cloudflare will publish the site at:

```
https://4dots.pages.dev/
```

The exact `pages.dev` hostname may differ if the project name is unavailable. After the first deployment, add the final hostname to Turnstile and the Apps Script `ALLOWED_HOSTNAMES` property, then set `RETURN_URL` to the full site URL.

For an authenticated manual deployment with Wrangler:

```
npx wrangler pages deploy docs
```

If the Cloudflare build configuration requires a deploy command, set it to:

```
npx wrangler pages deploy docs
```

Do not use `npx wrangler deploy`. That command deploys Workers and will reject the Pages-only `pages_build_output_dir` configuration.

Troubleshoot the Wrong Deploy Command

This error means the project was configured as Workers Builds instead of Pages, or its deploy command was overridden:

```
It seems that you have run `wrangler deploy` on a Pages project
Missing entry-point to Worker script or to assets directory
```

In Cloudflare, either create/connect the repository as a Pages project, or change the configured deploy command from `npx wrangler deploy` to `npx wrangler pages deploy docs`.

Security Notes

- Never commit API secrets, Sheet credentials, or Turnstile secrets.

- Keep the Google Sheet private and limit its editors.
- Use `ALLOWED_HOSTNAMES` in Apps Script for production.
- The receiver verifies Turnstile, validates allowed choices, limits field lengths, rejects malformed fields, uses a honeypot, locks concurrent writes, and neutralizes spreadsheet formula injection.
- Cloudflare Pages is appropriate for public lead forms, not passwords, card details, or other sensitive transactions.

Production Checklist

- Confirm `docs/app.js` contains the active Apps Script `/exec` URL and the public site key from the matching Turnstile widget.
- Submit one valid test lead and confirm it appears in the private Sheet.
- Confirm the sheet uses the current 11-column schema with no `Message` column.
- Test invalid email, invalid phone, missing consent, and no selected service.
- Test on mobile and desktop.
- Set `RETURN_URL` and `ALLOWED_HOSTNAMES` to the final Cloudflare Pages or custom-domain hostname.
- Confirm Cloudflare Pages serves the security headers from `docs/_headers`.
- After changing `Code.gs`, redeploy the existing Apps Script deployment as a new version.
- Add a privacy policy before running paid acquisition campaigns.

License

See [LICENSE](#).